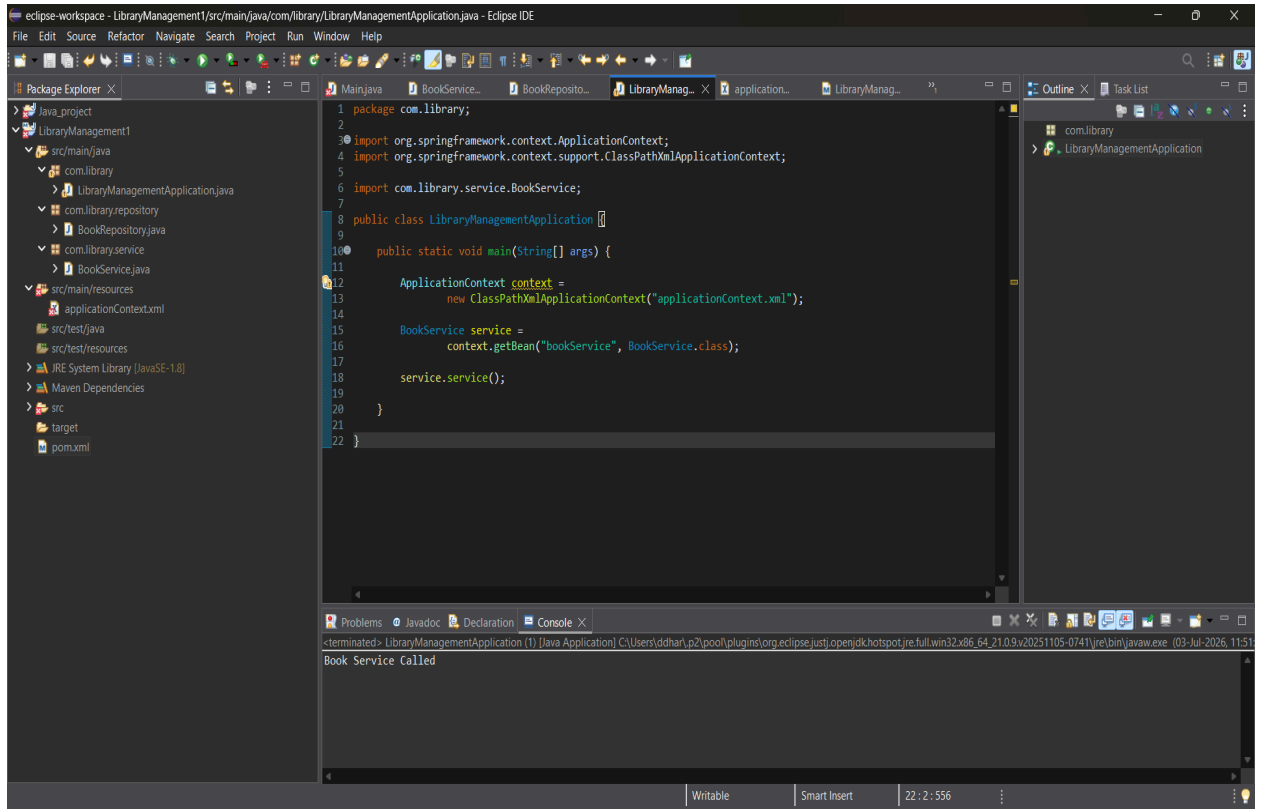
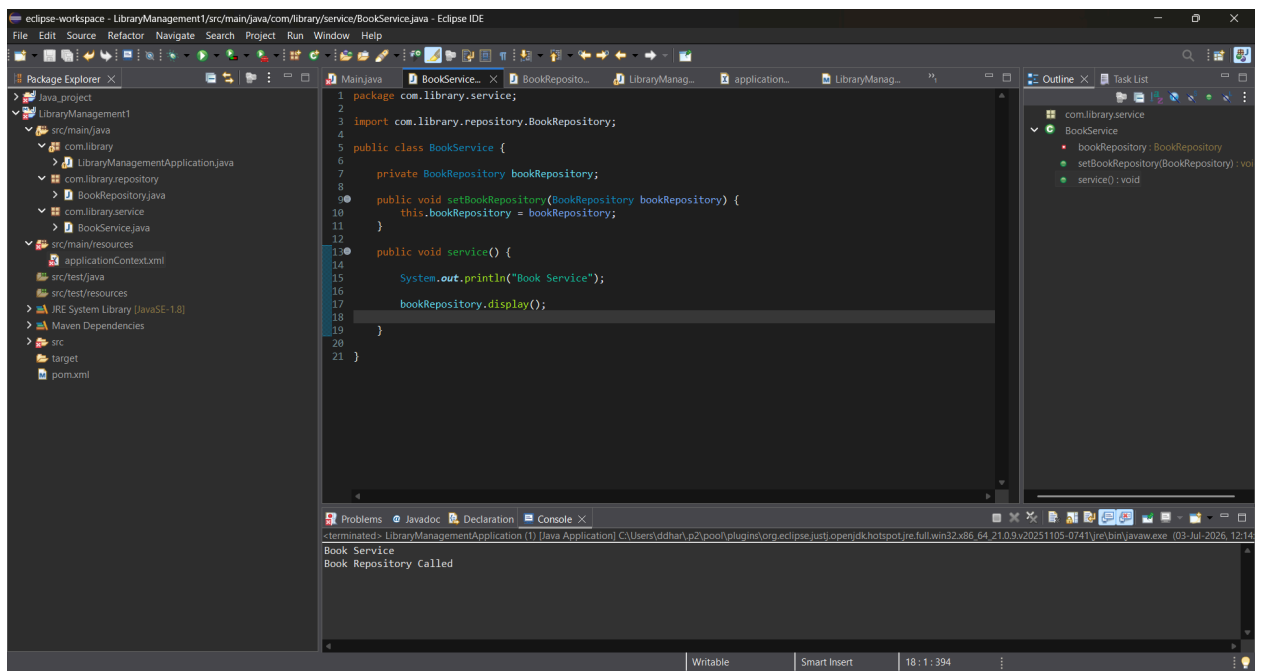




1.



2.

applicationContext.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans

```

```

https://www.springframework.org/schema/beans/spring-beans.xsd">
<bean id="bookRepository"
    class="com.library.repository.BookRepository"/>
<bean id="bookService"
    class="com.library.service.BookService">
    <property name="bookRepository" ref="bookRepository"/>
</bean>
</beans>

```

LibraryManagementApplication.java

```

package com.library;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;
public class LibraryManagementApplication {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
        BookService service =
            context.getBean("bookService", BookService.class);
        service.service();
    }
}

```

BookService

```

package com.library.service;
import com.library.repository.BookRepository;
public class BookService {
    private BookRepository bookRepository;
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }
    public void service() {
        System.out.println("Book Service");
        bookRepository.display();
    }
}

```

BookRepository

```

package com.library.repository;
public class BookRepository {
    public void display() {
        System.out.println("Book Repository Called");
    }
}

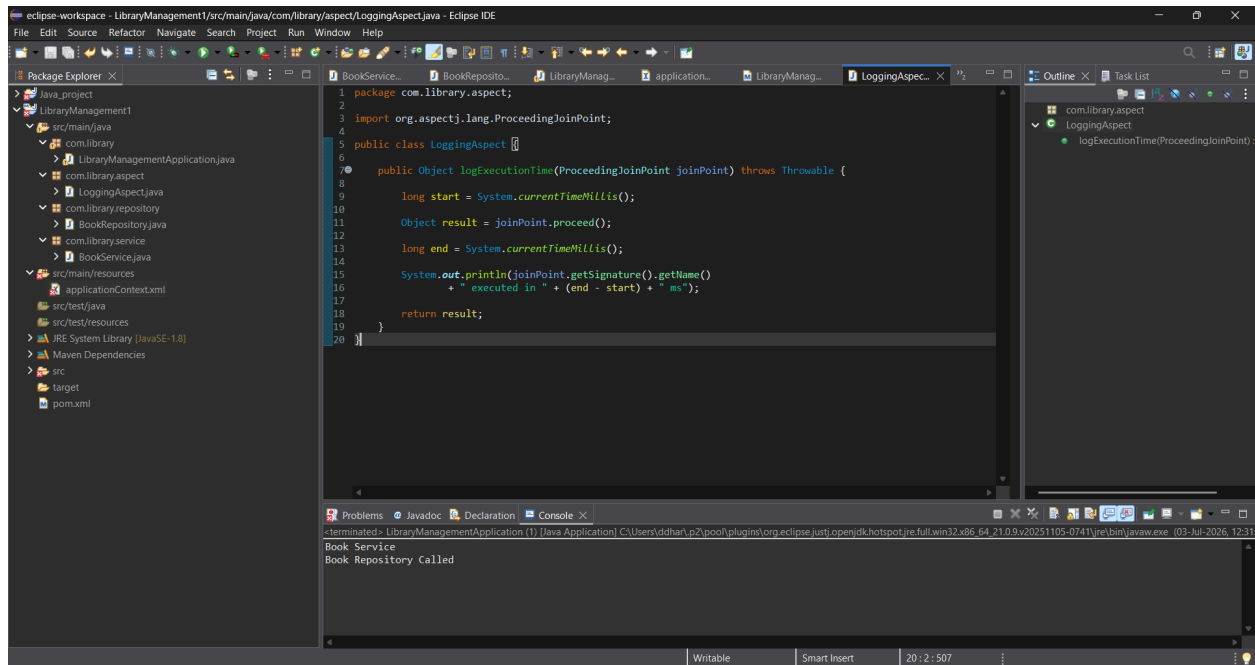
```

3. Aspect → LoggingAspect.java

```
package com.library.aspect;
import org.aspectj.lang.ProceedingJoinPoint;
public class LoggingAspect {
    public Object logExecutionTime(ProceedingJoinPoint joinPoint) throws Throwable {
        long start = System.currentTimeMillis();
        Object obj = joinPoint.proceed();
        long end = System.currentTimeMillis();
        System.out.println(joinPoint.getSignature().getName()
            + " executed in " + (end - start) + " ms");
        return obj;
    }
}
```

Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.library</groupId>
    <artifactId>LibraryManagement</artifactId>
    <version>1.0</version>
    <dependencies>
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-aop</artifactId>
        <version>5.3.30</version>
    </dependency>
    <dependency>
        <groupId>org.aspectj</groupId>
        <artifactId>aspectjweaver</artifactId>
        <version>1.9.21</version>
    </dependency>
    </dependencies>
</project>
```



```
4. <dependencies>
5. <dependency>
6. <groupId>org.springframework</groupId>
7. <artifactId>spring-context</artifactId>
8. <version>5.3.30</version>
9. </dependency>
10. <dependency>
11. <groupId>org.springframework</groupId>
12. <artifactId>spring-aop</artifactId>
13. <version>5.3.30</version>
14. </dependency>
15. <dependency>
16. <groupId>org.springframework</groupId>
17. <artifactId>spring-webmvc</artifactId>
18. <version>5.3.30</version>
19. </dependency>
20. </dependencies>
21. pom.xml
```

5. Spring IoC manages

- BookService Bean
- BookRepository Bean

through

```
<bean id="bookRepository"
      class="com.library.repository.BookRepository"/>

<bean id="bookService"
      class="com.library.service.BookService">

<property name="bookRepository"
          ref="bookRepository"/>

</bean>
```

6. BookRepository.java

```
package com.library.repository;

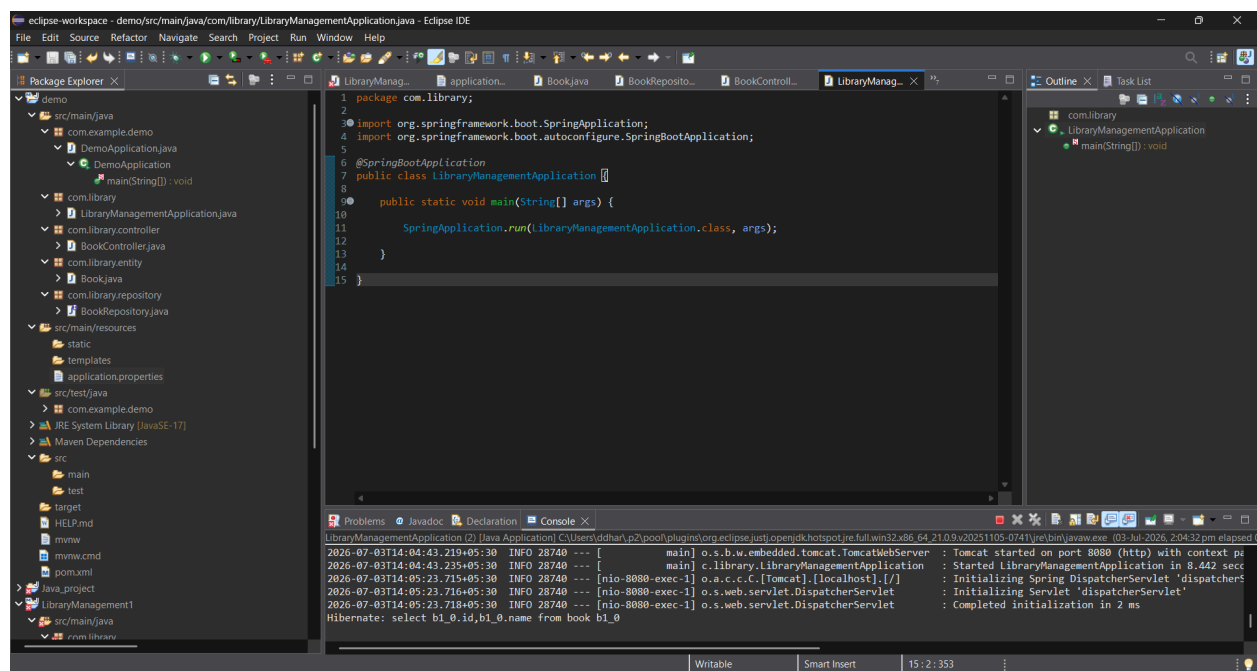
import org.springframework.stereotype.Repository;

@Repository
public class BookRepository {

    public void display() {
        System.out.println("Repository");
    }

}
```

9.



BookController.java

```
package com.library.controller;
```

```
import java.util.List;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.web.bind.annotation.*;
```

```
import com.library.entity.Book;
```

```
import com.library.repository.BookRepository;
```

```
@RestController
@RequestMapping("/books")
public class BookController {

    @Autowired
    private BookRepository repository;

    @PostMapping
    public Book addBook(@RequestBody Book book) {
        return repository.save(book);
    }

    @GetMapping
    public List<Book> getBooks() {
        return repository.findAll();
    }
}
```